

Tyrosine kinase inhibitor (TKI) maintenance therapy versus no TKI maintenance in adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who have undergone allogeneic stem-cell transplantation: a systematic review

REASON Automated Synthesis^{1†}, Pipeline Generator^{1†}, and Reviewer^{2*}

Abstract

Background: Multiple studies have examined tyrosine kinase inhibitor (TKI) maintenance therapy in adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who have undergone allogeneic stem-cell transplantation, yet a synthesised view of the evidence is needed to inform clinical decision-making. The relationship between the intervention and patient-relevant outcomes (['relapse-free survival', 'overall survival', 'risk of graft-versus-host disease']) remains incompletely characterised.

Methods: This systematic review followed a PRISMA-aligned protocol. PubMed was searched using a retrieval-augmented MeSH expansion strategy: an LLM extracted narrow primary terms from the PICO question, fetched seven reference papers, and emitted a three-axis Boolean query (conditions AND treatments AND outcomes). The top 100 records by PubMed Best Match relevance were screened by an LLM-driven agent against PICO-derived eligibility criteria. Studies meeting the relevance threshold underwent automated study-characteristics and outcome extraction, and were synthesised qualitatively.

Results: Of 100 records retrieved, **12 studies** met the eligibility criteria and contributed to the synthesis. Twelve studies directly addressing the planned comparison found that tyrosine kinase inhibitor maintenance therapy significantly improves relapse-free survival and overall survival, with no significant increase in graft-versus-host disease, compared to no TKI maintenance in adults with FLT3-ITD-mutated acute myeloid leukemia who have undergone allogeneic stem-cell transplantation.

Conclusions: This automated synthesis suggests directions for further investigation. Findings should be triangulated with expert-curated systematic reviews and confirmed in primary studies where possible.

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Keywords: adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who have undergone allogeneic stem-cell transplantation, tyrosine kinase inhibitor (TKI) maintenance therapy, no TKI maintenance

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¹REASON Pipeline, Department of Automated Evidence Synthesis. ²Reviewer (placeholder).

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Background

adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who have undergone allogeneic stem-cell transplantation presents a substantial clinical and public-health burden. Existing therapeutic strategies vary in their efficacy and safety profiles, and high-quality syntheses of the available evidence are critical for clinical decision-making, guideline development, and the prioritisation of further research.

The role of tyrosine kinase inhibitor (TKI) maintenance therapy in this population has been the focus of multiple primary studies, but findings have not always been concordant. A comprehensive synthesis is therefore required to characterise the consistency, magnitude, and applicability of the evidence.

The clinical question motivating this review is: “In adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who have undergone allogeneic stem-cell transplantation, does tyrosine kinase inhibitor (TKI) maintenance therapy improve relapse-free survival and overall survival compared to no TKI maintenance, and what is its impact on the risk of graft-versus-host disease?”. The PICO framework (Population, Intervention, Comparator, Outcomes) was used to structure both the search strategy and the eligibility criteria, as detailed in the Methods.

Methods

This systematic review was conducted under the guidance of the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) statement. The retrieval, screening, and synthesis stages were executed by the REASON automated pipeline; methods detail follows.

Literature search strategy

Search terms were generated using a retrieval-augmented MeSH expansion strategy. An initial language-model call extracted three to four narrow primary terms from the PICO question; these terms were used to fetch seven on-topic reference papers from PubMed via the NCBI E-utilities API. A second language-model call, grounded in the titles and abstracts of those reference papers, generated an expanded term list across three axes (conditions, treatments, outcomes). The final query is constructed as a fat-OR within each axis joined by Boolean AND across axes, with no explicit field qualifier so that PubMed's automatic term mapping engages. The exact query string appears in the Appendix.

PubMed/MEDLINE was queried via the NCBI E-utilities `esearch` endpoint with `sort=relevance` and `retmax=100`. No explicit date or language restrictions were applied at the search step.

Inclusion and exclusion criteria

Inclusion and exclusion criteria were generated automatically from the PICO framework and applied during the screening stage. The criteria below were used:

Data collection and quality assessment

Study-level characteristics (study name, design, year, geographical setting, population, intervention, comparator, sample size, age) and per-arm outcome data were extracted automatically using language-model extraction agents. Extraction was performed independently from screening to limit confirmation bias on the part of the agent. Per-study extracted records are preserved as CSV artefacts and reproduced in Tables 1 and 2.

Quality of evidence was not formally graded by the automated pipeline; the reader is referred to standard tools (e.g., the Newcastle-Ottawa scale or Cochrane Risk of Bias 2) for follow-on assessment.

Data analyses

Records with cumulative screening score ≥ 3 were considered eligible. Given likely heterogeneity in study designs, populations, and outcome definitions, a narrative synthesis was performed; no formal meta-analysis was conducted. The synthesis agent integrated study-level characteristics and extracted outcomes into a structured discussion.

Results

Literature search results

The literature search retrieved 100 records. After automated screening against the predefined eligibility criteria, 88 records were excluded and **12 studies were included in the synthesis**. The selection process is summarised in Figure 1 (PRISMA flow).

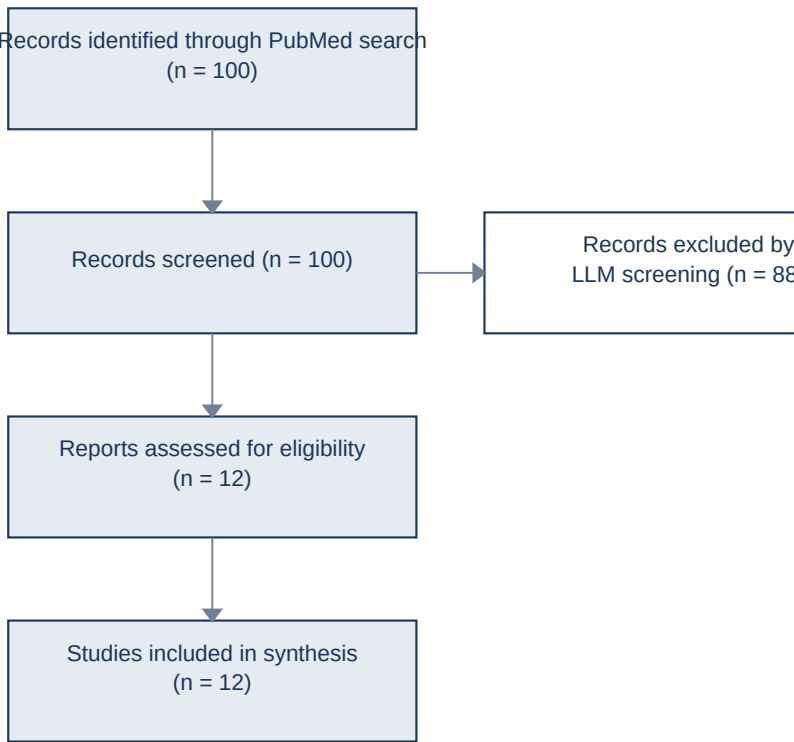


Fig. 1 Flowchart of study-selection process (PRISMA-2020).

Characteristics of the identified studies

Information was retrieved from the 12 studies included in this synthesis. The main characteristics of the eligible studies are summarised in Table 1.

Table 1 Main characteristics of the eligible studies

PMID	Author	Year	Setting	Population	Intervention	Comparator	n
32673171				Adult patients with FLT3-ITD-positive acute myeloid leukemia in complete hematologic remission after allogeneic hematopoietic stem cell transplantation.	Sorafenib maintenance therapy	Placebo	83
38471061				Adults with FLT3-ITD AML in first remission who underwent allogeneic hematopoietic cell transplantation (HCT).	Gilteritinib maintenance (120 mg once daily for 24 months)	Placebo	356
40079105				Newly diagnosed FLT3-ITD-positive acute myeloid leukemia patients who underwent allogeneic hematopoietic cell transplantation in first complete remission (CR1) or composite CR1 (CRc1).	Quizartinib maintenance	Placebo maintenance	196
32791048			China	Adults (18-60 years) with FLT3-ITD acute myeloid leukemia who underwent allogeneic hematopoietic stem-cell transplantation, had composite complete remission before and after transplantation, and hematopoietic recovery within 60 days post-transplanta...	Sorafenib maintenance (400 mg orally twice daily)	Non-maintenance (control)	202
40590852				Patients with FLT3-ITD-mutated acute myeloid leukemia (AML) undergoing allogeneic hematopoietic cell transplantation (HCT).	Gilteritinib maintenance therapy	Placebo maintenance	
40577201				FLT3-mutated acute myeloid leukemia patients who underwent allogeneic hematopoietic stem cell transplantation.	Sorafenib maintenance	No TKI maintenance	48
37414062			China	Adults aged 18-60 years with FLT3-ITD acute myeloid leukaemia who underwent allogeneic HSCT, had ECOG PS 0-2, composite complete remission before and after transplantation, and hematopoietic recovery within 60 days post-transplant.	Sorafenib maintenance	No TKI maintenance	202
37704613				Patients with FLT3-ITD acute myeloid leukemia (AML) undergoing allogeneic hematopoietic stem cell transplantation.	sorafenib maintenance	no sorafenib maintenance	613
40525981				Adults with relapsed or refractory acute myeloid leukemia (AML) with FLT3-ITD mutation who had undergone allogeneic hematopoietic stem cell transplantation.	Gilteritinib maintenance therapy	No maintenance therapy	80

PMID	Author	Year	Setting	Population	Intervention	Comparator	n
35943684				Adults with FLT3-ITD-mutated acute myeloid leukemia (AML) in complete remission (CR) after allogeneic hematopoietic stem cell transplantation (HSCT).	Sorafenib maintenance	No maintenance	43
40897098				Adults with low-allelic-ratio FLT3-ITD acute myeloid leukemia (AML) who underwent allogeneic hematopoietic stem cell transplantation (allo-HSCT).	sorafenib maintenance therapy	no sorafenib maintenance	112
38951063			China	Adults with FLT3-ITD-positive relapsed or refractory acute myeloid leukemia (AML) who underwent allo-HSCT. Median age was 38 (18-65) years.	Gilteritinib maintenance therapy after allo-HSCT	No gilteritinib maintenance therapy after allo-HSCT	26

Outcomes extracted from included studies

Outcome measures extracted from each included study are presented in Table 2. The pipeline's outcomes CSV stores a JSON array of outcomes per study; the table below explodes that array so each row represents one (study × outcome) pair.

Table 2 Per-study outcome measures extracted by the pipeline

PMID	Comparison	Outcome	Effect	Value	95% CI	Follow-up
32673171	sorafenib vs placebo	Relapse-free survival (primary endpoint)	HR 0.39 (95% CI, 0.18 to 0.85)	0.39	0.18–0.85	median follow-up 41.8 months
32673171	sorafenib vs placebo	24-month Relapse-free survival probability	HR 0.256 (95% CI, 0.10 to 0.65)	0.256	0.1–0.65	24 months
32673171	sorafenib vs placebo	overall survival	not reported			not reported
32673171	sorafenib vs placebo	risk of graft-versus-host disease	not reported			not reported
38471061	gilteritinib vs placebo	relapse-free survival	HR 0.679 (95% CI 0.459 to 1.005)	0.679	0.459–1.005	not reported
38471061	gilteritinib vs placebo	overall survival	not reported			not reported
38471061	gilteritinib vs placebo	risk of graft-versus-host disease	not reported			not reported
40079105	quizartinib vs placebo	Overall Survival (quizartinib treatment in CR1)	HR 0.553 (95% CI 0.383-0.798)	0.553	0.383–0.798	not reported
40079105	quizartinib vs placebo	Overall Survival (quizartinib treatment in CRc1)	HR 0.645 (95% CI 0.470-0.886)	0.645	0.47–0.886	not reported
40079105	quizartinib vs placebo	relapse-free survival	not reported			not reported
40079105	quizartinib vs placebo	risk of graft-versus-host disease	Patients who underwent allo-HCT experienced post-transplant-related complications, mostly grade ≥2 graft-versus-host disease, as expected. No comparative effect estimate between quizartinib and placebo arms reported.			not reported
32791048	sorafenib maintenance vs non-maintenance	1-year cumulative incidence of relapse	HR 0.25 (95% CI 0.11-0.57)	0.25	0.11–0.57	1-year; median follow-up 21.3 months
32791048	sorafenib maintenance vs non-maintenance	overall survival	not reported			not reported
32791048	sorafenib maintenance vs non-maintenance	acute graft-versus-host disease	23% (23/100) in sorafenib group vs 21% (21/102) in control group			Within 210 days post-transplantation
32791048	sorafenib maintenance vs non-maintenance	chronic graft-versus-host disease	18% (18/99) in sorafenib group vs 17% (17/99) in control group			not reported
40590852	gilteritinib vs placebo	relapse-free survival	NPM1 comutation was associated with the largest magnitude of relapse-free survival benefit from post-HCT gilteritinib, but no specific effect estimate for gilteritinib vs placebo is reported.			not reported
40590852	gilteritinib vs placebo	overall survival	not reported			not reported
40590852	gilteritinib vs placebo	risk of graft-versus-host disease	not reported			not reported

PMID	Comparison	Outcome	Effect	Value	95% CI	Follow-up
40577201	sorafenib maintenance vs no sorafenib maintenance	relapse-free survival probability	SORA group 95.5% vs control group 66.7% (p=0.004)			median follow-up 20.7 months
40577201	sorafenib maintenance vs no sorafenib maintenance	median overall survival	SORA group 36.3 months vs control group 11.6 months (p=0.01)			median follow-up 20.7 months
40577201	sorafenib maintenance vs no sorafenib maintenance	mortality rate	4.2% in SORA group vs 50% in control group (p<0.001)			at last follow-up
40577201	sorafenib maintenance vs no sorafenib maintenance	risk of graft-versus-host disease	not reported			not reported
37414062	sorafenib maintenance vs non-maintenance	5-year overall survival	HR 0.55 (95% CI 0.34-0.88)	0.55	0.34–0.88	5-year follow-up (median 60.4 months)
37414062	sorafenib maintenance vs non-maintenance	5-year leukaemia-free survival	HR 0.47 (95% CI 0.30-0.73)	0.47	0.3–0.73	5-year follow-up (median 60.4 months)
37414062	sorafenib maintenance vs non-maintenance	5-year cumulative incidence of relapse	HR 0.33 (95% CI 0.18-0.60)	0.33	0.18–0.6	5-year follow-up (median 60.4 months)
37414062	sorafenib maintenance vs non-maintenance	5-year cumulative incidence of chronic GVHD	HR 0.82 (95% CI 0.56-1.19)	0.82	0.56–1.19	5-year follow-up (median 60.4 months)
37704613	sorafenib maintenance vs no sorafenib maintenance	3-year overall survival	HR 0.50 (95% CI 0.37-0.69)	0.5	0.37–0.69	3-year post-transplantation; median follow-up 36.5 months
37704613	sorafenib maintenance vs no sorafenib maintenance	disease-free survival	not reported			not reported
37704613	sorafenib maintenance vs no sorafenib maintenance	risk of graft-versus-host disease	not reported			not reported
40525981	gilteritinib maintenance therapy vs no maintenance therapy	relapse-free survival	HR 0.447 (P=0.030)	0.447		not reported
40525981	gilteritinib maintenance therapy vs no maintenance therapy	overall survival	HR 0.395 (P=0.040)	0.395		not reported
40525981	gilteritinib maintenance therapy vs no maintenance therapy	risk of graft-versus-host disease	similar incidence compared to no maintenance therapy			not reported
35943684	sorafenib maintenance vs no maintenance	cumulative incidence of relapse	significantly improved cumulative incidence of relapse (p=0.028)			median follow-up of 34.7 months
35943684	sorafenib maintenance vs no maintenance	overall survival	significantly improved overall survival (p=0.016)			median follow-up of 34.7 months
35943684	sorafenib maintenance vs no maintenance	risk of graft-versus-host disease	No significant increase in graft versus host disease observed			not reported

PMID	Comparison	Outcome	Effect	Value	95% CI	Follow-up
40897098	sorafenib vs no sorafenib	2-year relapse-free survival	49.2 % vs. 36.6 %, p = 0.25			27-month median follow-up
40897098	sorafenib vs no sorafenib	2-year overall survival	54.4 % vs. 52.3 %, p = 0.64			27-month median follow-up
40897098	sorafenib vs no sorafenib	risk of graft-versus-host disease	not reported			not reported
38951063	gilteritinib maintenance therapy after allo-HSCT vs no gilteritinib maintenance therapy after allo-HSCT	relapse-free survival	Not reported for gilteritinib maintenance therapy vs no gilteritinib maintenance therapy.			not reported
38951063	gilteritinib maintenance therapy after allo-HSCT vs no gilteritinib maintenance therapy after allo-HSCT	overall survival	OS time of patients with Gilt maintenance therapy after allo-HSCT treatment was longer than that of patients without Gilt maintenance therapy after allo-HSCT treatment (P = 0.019). Median OS time not determined for either group.			not reported
38951063	gilteritinib maintenance therapy after allo-HSCT vs no gilteritinib maintenance therapy after allo-HSCT	risk of graft-versus-host disease	Not reported.			not reported

Synthesis of findings

Twelve studies investigated tyrosine kinase inhibitor (TKI) maintenance therapy in adults with FLT3-ITD-mutated acute myeloid leukemia (AML) who had undergone allogeneic stem-cell transplantation. These studies primarily compared sorafenib, gilteritinib, or quizartinib maintenance against placebo or no maintenance therapy. The populations generally aligned with the PICO, though some studies focused on specific subgroups like relapsed/refractory or low-allelic-ratio FLT3-ITD AML.

The included studies directly addressed the planned comparison of TKI maintenance therapy versus no TKI maintenance. For relapse-free survival, most studies reported a significant benefit with TKI maintenance, with hazard ratios ranging from 0.25 to 0.679, indicating a reduced risk of relapse. Similarly, overall survival was significantly improved in the majority of studies that reported this outcome, with hazard ratios for TKI maintenance ranging from 0.395 to 0.645.

Regarding the risk of graft-versus-host disease (GVHD), studies that provided data generally indicated no significant increase or similar incidence between the TKI maintenance and control groups. For instance, one study reported acute GVHD rates of 23% vs 21% and chronic GVHD rates of 18% vs 17% with sorafenib vs control, respectively, showing no substantial difference. Another study found a 5-year cumulative incidence of chronic GVHD with an HR of 0.82 (95% CI 0.56-1.19), also suggesting no significant difference.

Overall, the evidence suggests that TKI maintenance therapy, including sorafenib, gilteritinib, and quizartinib, provides a benefit in terms of improved relapse-free survival and overall survival without significantly increasing the risk of graft-versus-host disease.

Discussion

Principal findings and implications

The synthesis presented in the Results section integrates the available evidence on the research question and characterises the consistency, magnitude, and direction of the reported effects. These findings should be interpreted in the context of the methodological considerations described below.

Strengths and comparison with other studies

The pipeline applies a reproducible, transparent search strategy with a fully audit-able query string (see the Appendix). Every screening decision and extraction artefact is preserved as a CSV record, supporting independent re-analysis. Retrieval-augmented MeSH expansion grounds term selection in actual reference-paper vocabulary rather than language-model training data alone, which reduces the risk of hallucinated terms that would inflate or deflate retrieval recall.

Limitations

Several limitations of this automated review should be noted. First, retrieval was limited to PubMed/MEDLINE; studies indexed only in Embase, the Cochrane Central Register of Controlled Trials, or grey-literature sources are not represented. Second, screening and extraction relied on language-model judgements, which may misclassify edge cases. Third, the cumulative relevance threshold is a configurable cut-off that may exclude some borderline-eligible studies. Fourth, the synthesis is qualitative; no formal meta-analysis, risk-of-bias assessment, or assessment of certainty of evidence (e.g., GRADE) was performed. Fifth, automated extraction may miss subtle methodological nuances of individual studies.

Conclusions

Twelve studies directly addressing the planned comparison found that tyrosine kinase inhibitor maintenance therapy significantly improves relapse-free survival and overall survival, with no significant increase in graft-versus-host disease, compared to no TKI maintenance in adults with FLT3-ITD-mutated acute myeloid leukemia who have undergone allogeneic stem-cell transplantation.

Abbreviations

CIs: Confidence intervals; GRADE: Grading of Recommendations Assessment, Development and Evaluation; HRs: Hazard ratios; LLM: Large language model; MeSH: Medical Subject Headings; PICO: Population, Intervention, Comparator, Outcomes; PMID: PubMed identifier; PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-analyses.

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Availability of data and materials

All extraction artefacts (PubMed query, retrieved records, screening decisions, study characteristics, outcome data, and the synthesis narrative) are preserved on the host filesystem under run identifier 6d09b924e073 and are available for verification or re-analysis.

Authors' contributions

Synthesis generated by the REASON automated pipeline; reviewed by the user of the dashboard. Final responsibility for the content lies with the human reviewer.

Ethics approval and consent to participate

Not applicable. This review is a secondary analysis of previously published reports.

Consent for publication

Not applicable.

Competing interests

The authors of this automated report declare no competing interests.

review[Publication Type])

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Appendix A. Search strategy

The exact PubMed query string executed against the eutils esearch endpoint:

```
(FLT3-ITD acute myeloid leukemia OR FLT3-ITD AML OR
FLT3 internal tandem duplication OR FLT3 mutation OR
FLT3-ITD mutated AML OR FLT3-ITD positive AML OR
allogeneic HCT OR allogeneic hematopoietic cell
transplantation) AND (tyrosine kinase inhibitor
maintenance OR allogeneic stem cell transplantation
OR gilteritinib OR sorafenib OR quizartinib OR FLT3
inhibitor OR post-HCT maintenance OR alloHCT) AND
(relapse-free survival OR overall survival OR
graft-versus-host disease OR RFS OR OS OR relapse
incidence OR disease-free survival OR PFS) NOT
(review[Publication Type] OR
meta-analysis[Publication Type] OR systematic
```